

Database noSQL

Irene Siragusa, PhD

Outline

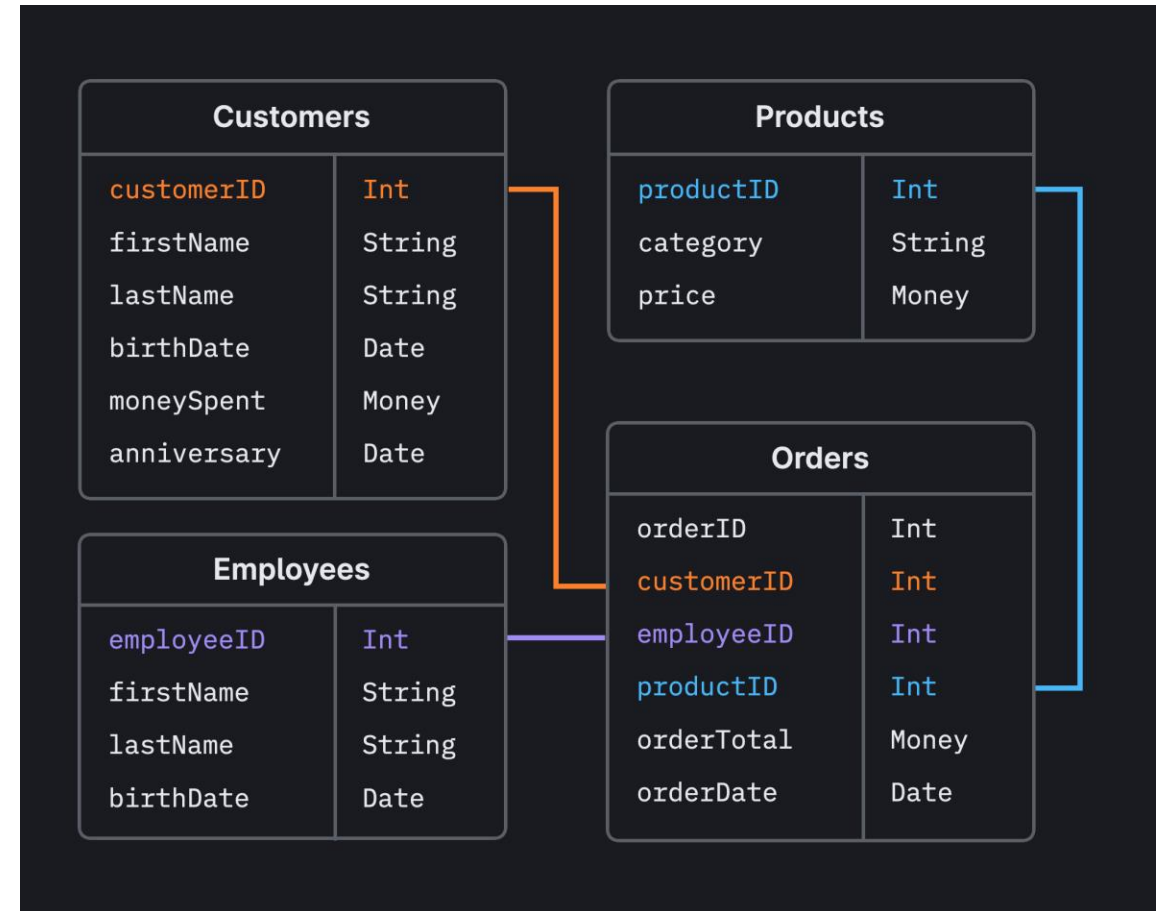
-  Database introduction
-  Relational Database
-  NoSQL database
 -  characteristics
 -  typologies
 -  pros and cons

Database introduction

- A database (DB) is an organized collection of data of interest for a particular application and it is managed by a Database Management System (DBMS).
- Relational databases organize data into tables (relations) composed of rows (records) and columns (attributes) with predefined data types.
- A relational database relies on a schema, which defines the structure, relationships, and constraints of the stored data and is queried and managed using Structured Query Language (SQL).

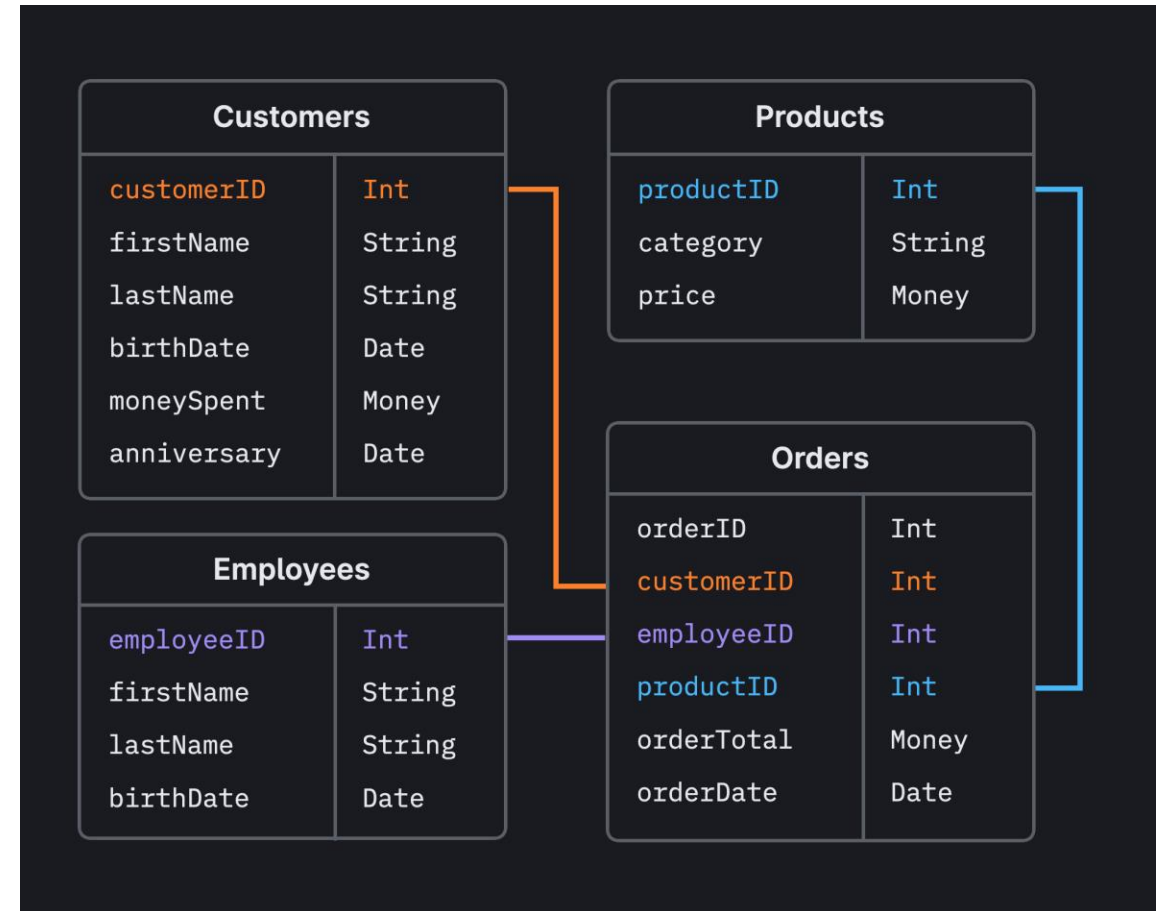
Relational Database

- Relational DBMS
- Relational Algebra
- Relational schema is created for the specific application
- Data constraints
- Primary and Foreign keys



Relational Database

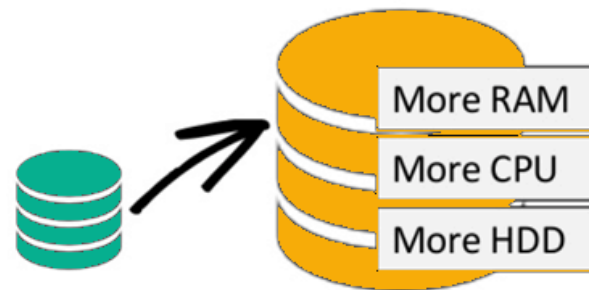
- Transactions
 - Group of operation
- ACID properties
 - Atomicity
 - Cannot be divided
 - Consistency
 - Meaningful
 - Isolation
 - Transactions' independency
 - Durability
 - Persistence



Big Data and Relational Database

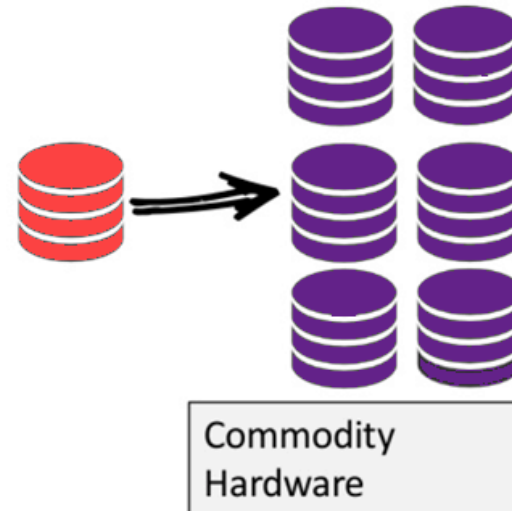
- When a company has a large quantity of data, RDBMS should be adjusted and scaled accordingly

Scale-Up (*vertical scaling*):



- Costly upgrading in hardware components
- Limited to improving a single machine or cluster

Scale-Out (*horizontal scaling*):



- RDBMS are replicated, shared and distributed
- Increases the number of machines

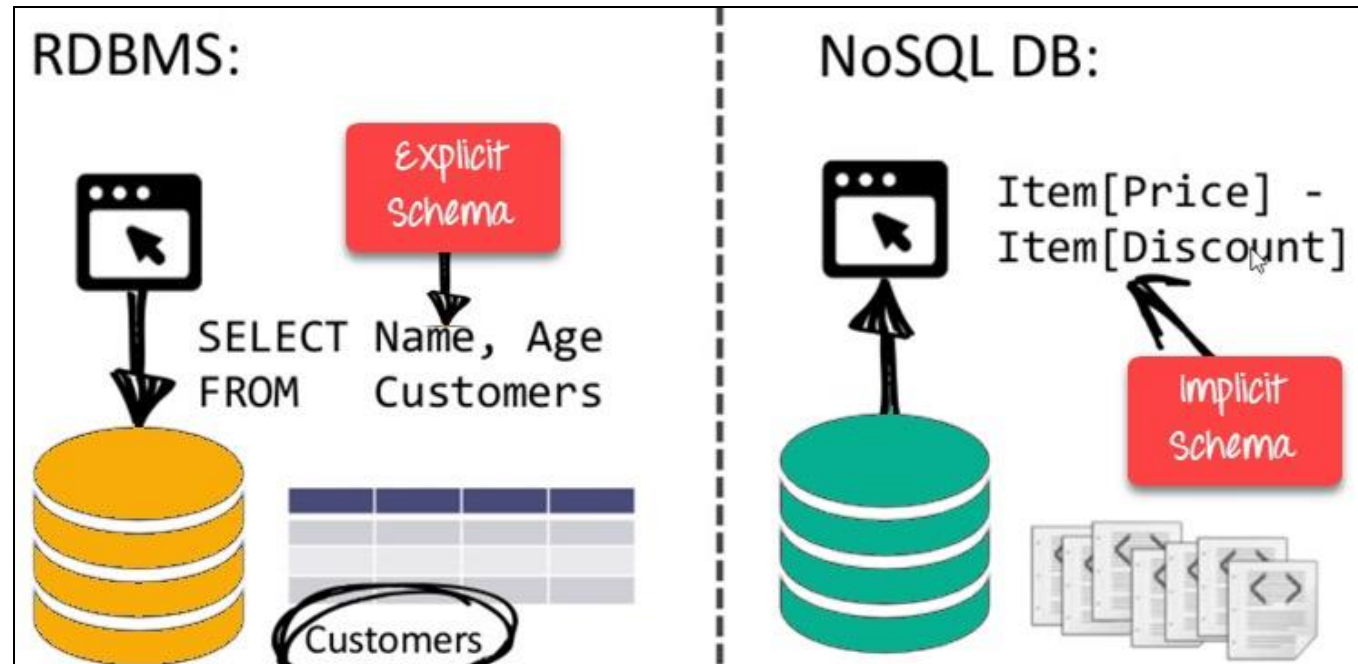
NoSQL database

- Not Only SQL
- No relational schema
- Tables have not a fixed format for its records
- Auto-consistent data structure (at record level) are used
- No query language
- No strict constraints in data types

NoSQL database – characteristics

Schema on write

- Apriori fixed schema



Schema on read

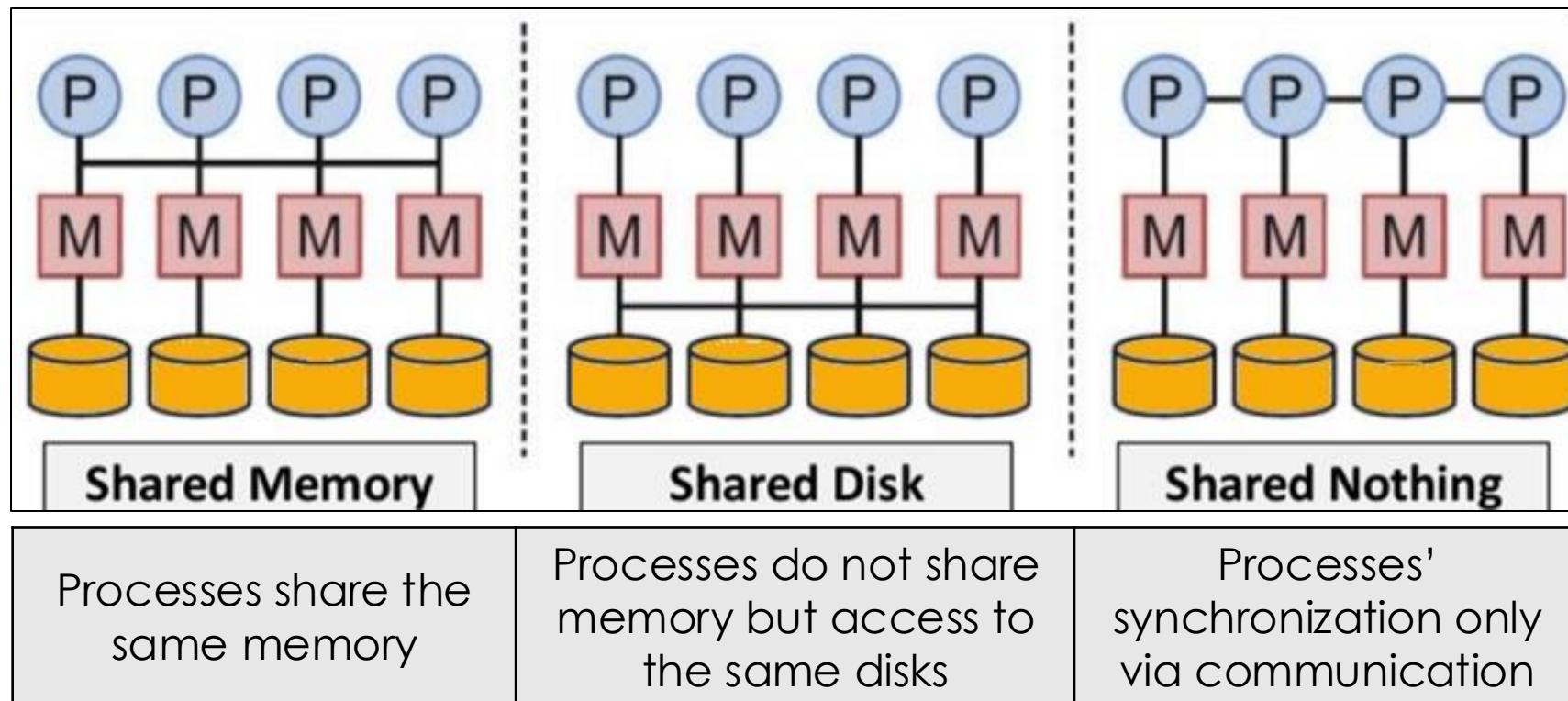
- Relaxed schema

NoSQL database – characteristics

- They offer simple Application Programming Interface (API) for their manipulation
- An HTTP REST interface is usually provided that can be accessed via textual protocols, eg JSON
- Web interface
- Distributed environment relying on eventual consistency
 - NoSQL databases do not have the concept of ACID transactions

NoSQL database – characteristics

- Shared nothing architecture

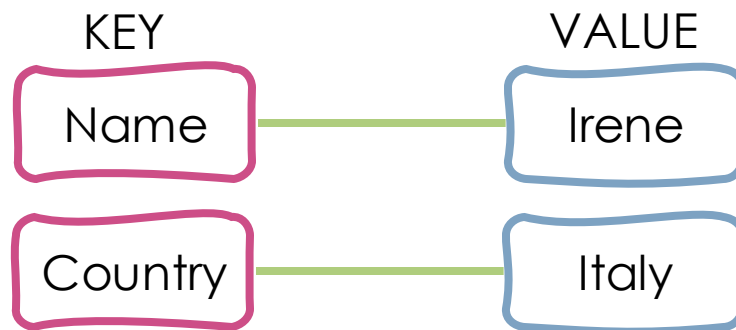


NoSQL database – characteristics

- No-relational data persistency
- Schemeless data
- High scalability
- Structured storage
 - Key-value pairs
 - Column oriented
 - Document oriented
 - Graph oriented

NoSQL database – typologies

- Key-value pairs
 - Simple data structure
 - Data are stored as key-value pairs, where
 - Key are unique
 - Heterogeneous values are allowed such as strings, JSON, BLOB (Binary Large Objects)
 - Key implicitly define the schema that is not required apriori
- Amazon Dynamo



NoSQL database – typologies

- Column oriented
 - Very efficient for cumulative queries (count, sum, average ...) along individual columns
 - Data are indexed directly on columns along which there are key-value pair
 - Can be grouped by “rows” or more generally in “column groups” that need not coincide with an entire record
 - Based on Google Big Table
 - Cassandra

ColumnFamily			
Row Key	Column Name		
	Key	Key	Key
	Value	Value	Value
	Column Name		
	Key	Key	Key
	Value	Value	Value

NoSQL database – typologies

- Column oriented

ROWID	Name	Birthday	Hobbies
1	Jos The Boss	11-12-1985	archery, conquering the world
2	Fritz von Braun	27-1-1978	building things, surfing
3	Freddy Stark		swordplay, lollygagging, archery
4	Delphine Thewiseone	16-9-1986	

Row-oriented lookup: from top to bottom and for every entry all columns are taken in memory.

Name	ROWID	Birthday	ROWID	Hobbies	ROWID
Jos The Boss	1	11-12-1985	1	archery	1, 3
Fritz Schneider	2	27-1-1978	2	conquering the world	1
Freddy Stark	3	16-9-1986	4	building things	2
Delphine Thewiseone	4			surfing	2
				swordplay	3
				lollygagging	3

A column-oriented database stores each column separately

NoSQL database – typologies

- Document oriented
 - Each record is a document in a structured format such as JSON and/or XML
 - Suitable for e-commerce, blogging platforms, analytics, ...
 - MongoDB



Relational data model

Highly-structured table organization with rigidly-defined data formats and record structure.

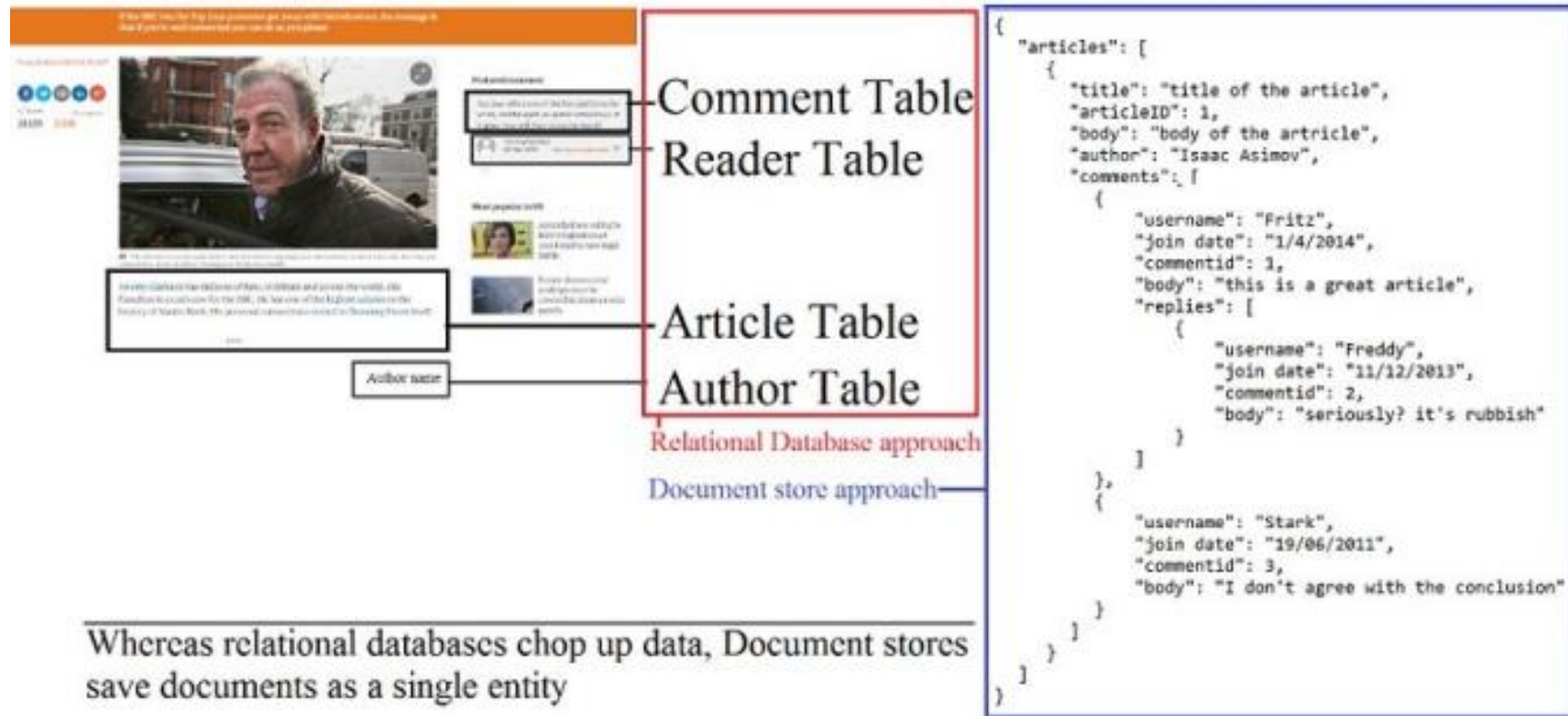


Document data model

Collection of complex documents with arbitrary, nested data formats and varying "record" format.

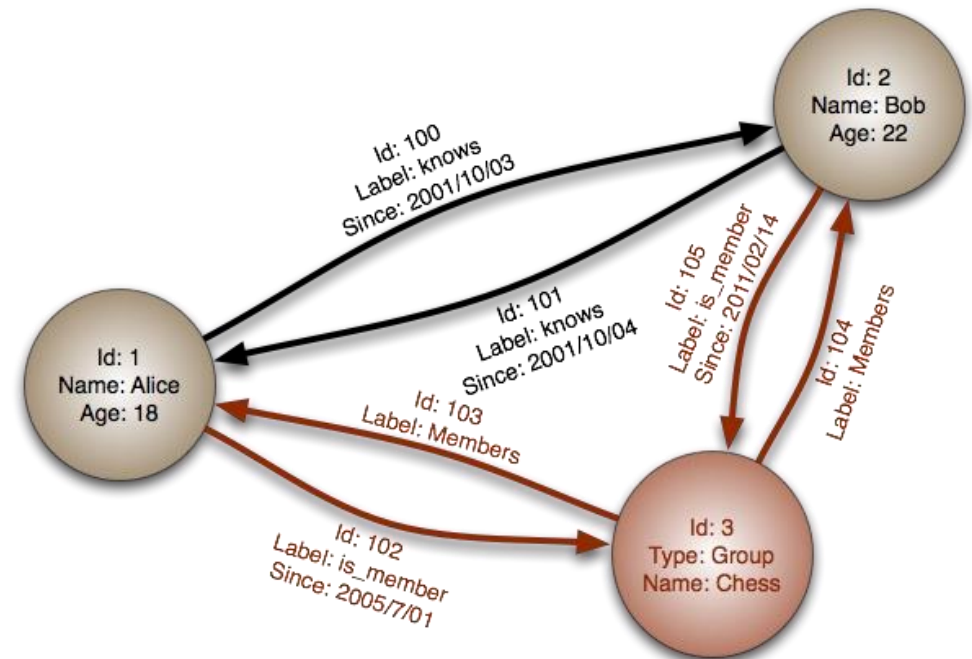
NoSQL database – typologies

- Document oriented



NoSQL database – typologies

- Graph oriented
 - Data are represented as labeled nodes and arcs of a graph
 - Relationships are modeled as archs
 - Multi-relational structure
 - Queries are graph operations (eg shortest path)
 - Neo4j



NoSQL database - pros and cons

- ✓ Can be used for both analytics purposes and as a data lake.
- ✓ Oriented to Big Data
- ✓ There is no longer a Single Point Of Failure (SPOF)
- ✓ Performance and horizontal scalability
- ✓ Structured, semi-structured or unstructured data
- ✓ Easy Object Oriented Programming (OOP)
- ✓ Does not require high-performance dedicated servers
- ✓ Designed for distributed databases
- ✓ Schemes on non-default data that can be altered without service interruption

NoSQL database - pros and cons

- ✗ No standardization
- ✗ Limited query capabilities
- ✗ RDBMS technologies are much more mature
- ✗ Open source systems are not highly popular in public or private organizations